

Joshua Max Manela

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EDUCATION

Michigan Technological University

Fall 2011 - Spring 2016

Graduated Magna Cum Laude BS Computer Engineering with a Minor in Computer Science, GPA: 3.74

WORK EXPERIENCE

Founding Engineer, Bedrock Robotics

May 2024 - Present

- Leads the ML efforts at Bedrock, resulting in a successful \$80MM oversubscribed series A round.
- Built a large portion of the initial ML and onboard infrastructure at the inception of the company.
- Regularly train, evaluate, and iterate on end-to-end behavior policies for heavy machinery tasks in pytorch.
- Design rust-based infrastructure for running models onboard the machine.
- Filed two patents related to novel goal-conditioning strategies for imitation learning for heavy machinery.

Staff Software Engineer, Perception, Waymo LLC (Formerly the Google Self-Driving Car Project)

January 2021 - May 2024

- Responsible for the development, integration, and maintenance of novel sensor fusion nets.
- Oversaw the technical work of roughly thirty to forty engineers at a time across hardware, software, and systems.
- Lead the consolidation of ten models into a single model on the car, running driverless across several cities.
- Published with the research team to push the state-of-the-art in long range sensor fusion.
- Lead the freeway radar perception working group composed of hardware, software, and systems engineers.
- Tech lead for long-range stereo-radar sensor fusion algorithm development.

Senior Software Engineer, Special Projects (Perception), Argo AI

March 2017 - January 2021

- Lead all aspects of the development and deployment of shipping a deep lidar object detection model.
- Designed, implemented, and integrated a deep stereo algorithm published at CVPR. (See Publications)
- Optimized a deep image segmentation system to run in real time and in production.
- Implemented hardware-level camera and lidar drivers which cross all levels of the stack.
- Designed and implemented a Gaussian Process based ground surface segmentation with cm-level precision.
- Optimized C++17 computer vision primitives to run faster than existing libraries.

Software Engineer, Machine Learning and Prediction, Uber Advanced Technology Group (ATG)

May 2016 - March 2017

- Worked with the motion planning team to develop deep-learning based controls algorithms.
- Wrote real-time visualizations for on-car debugging of our models.
- Architected onboard software architecture for long-horizon actor prediction.

Hardware Validation Engineering Intern, Facebook

Summer of 2015

- Wrote custom libraries to interface with the switches and run various network and stress tests.
- Designed and custom test harnesses for our custom network switches.
- Optimized network switch placement in Facebook data centers.

Software Development Engineering Intern, Amazon

August 2014 - November 2014

- Thorough preparation of software design with rigorous agile-centered design evaluations.
- Implemented RESTful APIs interfacing with a NoSQL database.
- Designed a service that validates security certificates between microservices across the Amazon stack.

Software and Controls Engineering Co-op, Mercury Marine

January 2013 - August 2013

- Extensively used MatLab and Simulink to write Engine Controller Unit (ECU) code.
- Soldered, crimped, designed, and built engine harness for production and test designs.

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- Wrote CAN, RFID, and IO libraries in C for an ATMEL AT90CAN128 processor for engine security firmware.

Simulation Engineering Intern, Applied Manufacturing Technologies

Summer 2012

- Laid out and simulated industrial robot performance using the Delmia I-GRIP CAD/Sim Package.
- Collaborated with other Simulation Engineers and wrote macros to improve CAD and simulation speed.

Lead Instructor, The Robot Garage

Summer 2011, 2012

- Wrote K-12 curriculum and led robotics classes and camps for the robot garage.
- Focused on the integration of mechanical and software design.
- Created the "Sumo Bot" stand-alone customizable robotics system still in use today.

Software Engineer, User Friendly EMR (Electronic Medical Record)

August 2013 - April 2014

- Developed various modules for an EMR system with the intent of organizing patient data.
- Designed a module to store employee signatures and credit card information in a MySQL database.

ACADEMIC EXPERIENCE

Undergraduate Researcher, Michigan Technological University

September 2013 - April 2016

- Designed and fabricated peripheral hardware kit for a robotic quadcopter to survey highways and bridges.
- Reconstructed a real-life bridge after having been flown over by a drone SLAM algorithm.
- Researched PSO (Particle Swarm Optimization) variants and published a novel meta-optimization for PSO.

Learning Center Coach, Michigan Technological University

May 2015 - May 2016

- Tutored students in both the Electrical Engineering and the Computer Science learning centers.
- Taught and tutored students in several classes including, Intro to Circuits I & II, Signal Processing, Electronics, Linear Systems, Intro to Computer Science, Data Structures, Discrete Structures, Data Structures, and any other course on the undergraduate catalog in either the ECE or CS departments.

Student Research Assistant, University of Miami

Summer of 2010

- Participated and helped analyze information for heart-rate and stress testing.
- Used classic computer vision algorithms to measure the radial dimensions of arteries from an ultrasound machine.

Patents and Publications

Hwang, J. J., Kretzschmar, H., **Manela, J.**, Rafferty, S., Armstrong-Crews, N.: CramNet: Camera-Radar Fusion with Ray-Constrained Cross-Attention for Robust 3D Object Detection. In: ECCV (2022)

Methods and systems for constructing map data using poisson surface reconstruction
HU Xiaoyan, M Happold, **JM Manela**, G Hotson - US Patent App. 16/722,641, 2021

G. Yang, **J. Manela**, M. Happold, D. Ramanan. Hierarchical Deep Stereo for High Res Imagery. Computer Vision and Pattern Recognition Conference 2019 (CVPR)

J. M. Manela and T. C. Havens, "Histogram particle swarm optimization (HistPSO): Evolving non-parametric acceleration distributions," 2016 IEEE Congress on Evolutionary Computation (CEC), Vancouver, BC, Canada, 2016, pp. 2071-2076, doi: 10.1109/CEC.2016.7744043.

Deilamsalehy, H., Havens, T. C., and **Manela, J.** (April 18, 2017). "Heterogeneous Multisensor Fusion for Mobile Platform Three-Dimensional Pose Estimation." ASME. J. Dyn. Sys., Meas., Control. July 2017; 139(7): 071002. <https://doi.org/10.1115/1.4035452>

Volunteer Work

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FIRST Robotics Mentor, FIRST Robotics Teams 27, 2194, and 2586

2011 - 2015

- Mentored three different high-school and two different middle school robotics teams.
- Taught students the basics of engineering design, application, and programming.
- Helped students create presentations and organize events to promote their team and gain funding.